

Weighted Graph Neural Additive Networks for Structural Connectomes: Interpretable by Design, Audited Out-of-Sample

Iakov Odesser – 209860188 Asaf Denish – 211883855 Adam Vogel – 318900255

Graph Machine Learning, Tel Aviv University

Abstract

Graph Neural Additive Networks (GNANs) use hop distance on binary graphs, whereas structural connectomes are dense and weighted. We introduce W-GNAN, which replaces hop-distance modulation with a monotone edge-weight function while retaining an exact algebraic decomposition of the logit for a fixed graph. On synthetic graphs with fixed topology, topology-only variants are at chance; weighted variants reach AUROC 0.87–0.95 and recover the generating weight-function shape over the observed support. On 869 structural connectomes (60 positive screens for probable PTSD), W-GNAN achieves AUROC 0.584 [0.52, 0.65] versus 0.588 [0.52, 0.65] for edge-wise logistic regression; the paired interval does not resolve a difference. A held-out region-level audit finds that none of three hypothesis-motivated networks exceeds its label-permutation null, while GNNExplainer node masks are stable but nearly label-invariant. Exact model decomposition therefore does not by itself establish a reproducible neurobiological marker.¹

1 Introduction

Diffusion MRI turns each brain into a weighted graph: regions are nodes and white-matter tracts, quantified by tractography-derived weights, are edges. PTSD has been linked to altered large-scale organisation of such networks (Suo et al., 2017; Akiki et al., 2017), making graph learning a natural tool for studying it. Explanations of predictions may be useful for model auditing, yet message-passing GNNs (Kipf and Welling, 2017; Veličković et al., 2018) commonly require post-hoc tools (Ying et al., 2019). GNAN (Bechler-Speicher et al., 2024) is algebraically decomposable, but lets nodes interact through their *hop distance*, which is uninformative in dense connectomes when nearly every

pair is one or two hops apart. Our cohort has only 60 positive screens, the small-sample regime typical of clinical neuroimaging. We therefore test the mechanism where it can be verified (synthetic data) and treat the real-data results, positive or negative, as an empirical study with correspondingly limited inferential scope.

Research questions. Methodologically, can a one-hop GNAN be extended to dense, continuously weighted graphs while retaining an exact fixed-graph logit decomposition? Empirically, in a pilot cohort of 869 subjects (60 PCL-5 positive screens), how does the extension compare with linear and GNN baselines for binary probable-PTSD classification, and do region-level explanations replicate on held-out subjects?

Contributions. (C1) W-GNAN, with region-specific shape functions and a monotone edge function $g(w)$, whose logit decomposes into node and pairwise-interaction terms and which reduces to a specified mean-aggregating one-hop GNAN on binary graphs (§3). (C2) A synthetic validation in which topology is uninformative by construction (§5.1). (C3) A benchmark on shared frozen splits with bootstrap CIs and selected-nuisance checks (§5.2–5.3). (C4) A region-level interpretability audit: a descriptive in-sample/held-out comparison, label-permutation tests of hypothesis-motivated networks, and intrinsic/post-hoc agreement (§5.4).

2 Related Work

Graph learning on connectomes. Message-passing GNNs (Kipf and Welling, 2017; Veličković et al., 2018) are the default model for connectomes, benchmarked across many datasets by NeuroGraph (Said et al., 2023). For structural connectomes, Kazi et al. (2024) combine edge- and node-centred convolutions in a multi-head GCN, and Jeong et al. (2025) report a pilot GNN study of trauma-exposed

¹Code for this project is available at <https://github.com/yodesser/W-GNAN-Connectomics>

youth. Such models are not additively decomposable in their usual form; post-hoc methods such as GNNExplainer (Ying et al., 2019) optimize masks rather than expose an algebraic prediction sum.

Interpretable-by-design models. Generalised additive models (Hastie and Tibshirani, 1986) write a prediction as a sum of univariate shape functions; neural additive models (Agarwal et al., 2021) learn them with small networks. GNAN (Bechler-Speicher et al., 2024) carries this to graphs: node features pass through shape functions and interact through a learned function of *hop distance*, making the model exactly decomposable. The distance mechanism is designed for binary topology. Our narrower contribution is a monotone weight-modulated, one-hop interaction with a fixed-graph logit decomposition; we do not claim the first weighted or interpretable graph model in general, as weighted (Kawahara et al., 2017) and interpretable (Li et al., 2021) connectome networks exist, though neither is additively decomposable.

PTSD connectomics and small-sample evaluation. Network accounts of PTSD motivate attention to default-mode, salience and central-executive systems (Menon, 2011; Akiki et al., 2017), and graph studies report disrupted structural organisation (Suo et al., 2017). Machine-learning studies classify PTSD from such networks (Nicholson et al., 2020), but the largest multisite effort finds only modest out-of-sample accuracy (Zhu et al., 2023) and data-driven markers remain unestablished (Neria et al., 2022). With few positives, cross-validated estimates carry large error bars (Varoquaux, 2018) and leakage inflates results (Kapoor and Narayanan, 2023), which motivates frozen splits, inner-fold selection, bootstrap intervals and a held-out audit. These prior findings motivate hypotheses; they are not ground-truth tract biomarkers for our cohort.

3 Methods

3.1 From GNAN to weighted graphs

In GNAN (Bechler-Speicher et al., 2024), node features pass through learned shape functions and nodes interact through a function of shortest-path distance. In a diagnostic on 20 evenly spaced subjects from our cohort, median density was 0.498 and over 99.99% of region pairs were one or two hops apart. Hop distance therefore has little resolution here, while binarisation discards continuous

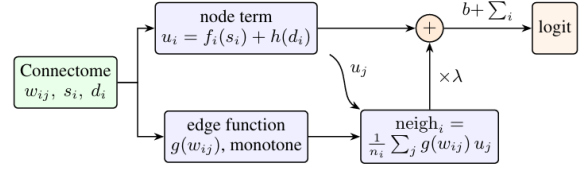


Figure 1: W-GNAN. Strength and degree are graph-derived. The pairwise quantity in Eq. 2 is an interaction contribution to the fixed-graph logit, not an edge-removal or causal effect.

weights.

For transformed matrix L , let $s_i = \sum_j L_{ij}$, let d_i be row-support degree divided by 431, and let $w_{ij} = \text{clip}(L_{ij}/q_{.99}^{\text{train}}, 0, 1)$. After train-fold standardisation, W-GNAN (Figure 1) forms $u_i = f_i(s_i) + h(d_i)$, with region-specific f_i and shared h :

$$\begin{aligned} \text{neigh}_i &= n_i^{-1} \sum_j m_{ij} g(w_{ij}) u_j, \\ n_i &= \max(1, \sum_j m_{ij}). \end{aligned} \quad (1)$$

where the implemented interaction mask is $m_{ij} = 1[L_{ij} > 0]$. The monotone piecewise-linear g has nine knots; positive softplus increments are cumulatively summed and normalized, enforcing $g(0) = 0, g(1) = 1$. It starts at the identity, while $\lambda = \text{softplus}(a) \geq 0$ starts at 1. Then $t_i = u_i + \lambda \text{neigh}_i$ and $\ell = b + \sum_i t_i$. Each f_i has seven knots on $[-3, 3]$, h has nine, and their knot values and the bias start at zero; all node shapes are anchored at zero input. No self-loops are supplied; an isolate has zero neighbour term. The binary-topology and fixed-linear ablations replace g by $1[w > 0]$ and w , respectively.

3.2 Exact fixed-graph decomposition

For a symmetric graph without self-loops, regrouping the directed neighbour sums gives

$$\begin{aligned} \ell &= b + \sum_i u_i + \lambda \sum_{i < j} C_{ij}, \\ C_{ij} &= m_{ij} g(w_{ij}) (u_j/n_i + u_i/n_j). \end{aligned} \quad (2)$$

This identity explains the model’s *logit for the observed fixed graph*. It is not a probability decomposition, intervention, or causal claim. Moreover, changing/removing an edge can change s_i and d_i , hence u_i ; C_{ij} is therefore the explicit interaction term, not the entire perturbation effect of tract (i, j) . The region allocation t_i and pairwise C_{ij} are distinct explanatory objects.

Property 1 (specified binary reference). If the supplied weights and mask agree and $w_{ij} \in \{0, 1\}$, the endpoint constraints make neigh_i the mean of u_j over one-hop neighbours. W-GNAN then equals the mean-aggregating, one-hop GNAN variant used here; this is not an equivalence claim for every GNAN configuration.

4 Experimental Setup

Cohort. The single-site cohort was provided by Yaniv Assaf’s laboratory. The available pool contained 1,191 scans from 919 participants. Density bounds $0.40 < d < 0.65$ removed 74 scans from 50 artifact-group participants, all of whose scans were excluded; selecting one scan per remaining participant yielded 869 unique subjects. Nodes are 400 Schaefer cortical parcels (Schaefer et al., 2018) plus 32 Tian-S2 subcortical parcels (Tian et al., 2020); matrices came from iFOD2 tractography with SIFT2 streamline weights (Smith et al., 2015). The target is a positive screen for *probable* PTSD, $y = 1[\text{PCL-5} \geq 33]$ (Blevins et al., 2015): 60/869 (6.9%). PCL-5 was complete (median 3, IQR 0–11); age was 31.1 ± 9.8 years, with 536 coded male, 332 female and one mode-imputed missing value. Acquisition descriptors were unavailable in the analysis artifacts.

Preprocessing. For each subject, with $T = \sum_{kl} W_{kl}$ and present-edge set E , the code computes $A = W/(T + 10^{-12})$ and $L_{ij} = \log_{10}(A_{ij} + 10^{-8}) - \min_{(k,l) \in E} \log_{10}(A_{kl} + 10^{-8})$ on E , and zero otherwise. Up to numerical offsets this is $\log_{10}(W_{ij}/\min_E W)$: T cancels, so weights are relative to the weakest present edge. That edge maps to zero and the tested W-GNAN mask $L > 0$ drops it, although raw-support degree retains it. W-GNAN scales L by the training-fold 99th percentile to $[0, 1]$; all learned transforms are fit on training data only.

Models. The reference is ℓ_2 logistic regression (LR) on 93,096 upper-triangle edges; LR strengths uses 432 row sums (balanced weights, $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$). Two-layer GAT/GCN models use the training-set 20% lowest-CV edges; node features are each L row, Yeo-7 one-hot and participation coefficient, followed by mean/max pooling, dropout .6 and class-weighted loss. Their results therefore describe this sparsification policy, not unrestricted GNN performance. W-GNAN uses AdamW (lr 5×10^{-3}), batch 64, at most 200

epochs, validation-AP stopping (patience 25), gradient norm 5, class weight n_0/n_1 , zero weight decay, and 10^{-3} penalties on mean-squared region-shape knots and second-difference smoothness of the node/gate shapes.

Prediction protocol. All models share stratified 5-fold CV repeated three times (seed 2026), with a stratified 20% inner validation split. Hyperparameters, stopping epoch and balanced-accuracy threshold use validation data; test folds are scored once. GNNs use repeat 0. Within each repeat, fold scores are percentile-ranked, pooled and scored; reported AUROC/AUPRC are means across repeats (AUPRC prevalence .069). A 1,000-sample clustered bootstrap resamples subject IDs and retains all repeated predictions; paired deltas use the same IDs. GNN deltas use only the matching repeat-0 LR predictions. These frozen-prediction intervals omit retraining/selection uncertainty. With 60 positives, an AUROC near .6 already has $\text{SE} \approx .04$ (Varoquaux, 2018).

Region-level audit. Each test fold yields a region score c_i . For W-GNAN, c_i is the held-out group difference in allocated totals $t_i = u_i + \lambda \text{neigh}_i$ (we do not audit C_{ij}); for LR, it is the coefficient times the standardised held-out group difference. A network’s score is $(|c_i| - \bar{c})/\text{sd}(c)$ within each fold, averaged over folds and over the network’s cortical Yeo-7 nodes (Yeo et al., 2011); “Cont” proxies Central Executive and subcortex is excluded. *Stability* is the mean pairwise Spearman correlation between the folds’ score vectors c . The three networks were proposal/literature-motivated, but the exact protocol came later. Global label permutations preserve total class count; transforms/W-GNAN are refit, while LR reuses observed-run fold-specific C (500 LR/15 folds; 100 W-GNAN/5 folds). One-sided $p = (1 + \#\{N \geq O\})/(P + 1)$ with $\alpha = .05/3$. As a descriptive reference, we also compare region rankings from an earlier W-GNAN prototype (different architecture and training population), refit with five seeds on all subjects and in 25 held-out cross-validation fits. GNNExplainer jointly optimizes nonnegative node and edge masks; we audit its node mask (100 epochs, positive-class target) for all 60 positives and 12 test negatives randomly selected per fold with seed 0, each with its held-out GAT.

Case	Topology	$g(w) = w$	Learned g	$\text{corr}(g, \hat{g})$
linear	0.50±0.00	0.92±0.05	0.93±0.03	0.996
threshold	0.50±0.00	0.91±0.01	0.95±0.04	0.982
saturating	0.50±0.00	0.87±0.03	0.88±0.05	0.969
binary	0.97±0.02	0.97±0.02	0.97±0.02	—
regions (node-only AUROC)		0.80±0.05		top-4 prec. 0.95

Table 1: Synthetic test AUROC (mean±SD, five seeds). $\text{corr}(g, \hat{g})$ is Pearson correlation over the 5th–95th percentile of observed training weights; last row: node-only W-GNAN AUROC and top-4 precision for planted regions.

Model	AUROC	AUPRC	Δ AUROC
Majority	0.500 [0.50, 0.50]	0.069 [0.05, 0.09]	−0.087 [−0.15, −0.02]
LR edges	0.588 [0.52, 0.65]	0.093 [0.07, 0.14]	—
LR strengths	0.572 [0.50, 0.64]	0.099 [0.07, 0.15]	−0.017 [−0.06, +0.03]
GAT*	0.539 [0.46, 0.62]	0.089 [0.06, 0.15]	−0.030 [−0.12, +0.07]
GCN*	0.515 [0.44, 0.60]	0.075 [0.05, 0.11]	−0.054 [−0.12, +0.01]
W-GNAN node-only	0.558 [0.51, 0.61]	0.083 [0.06, 0.11]	−0.029 [−0.09, +0.03]
W-GNAN binary topo.	0.552 [0.49, 0.61]	0.088 [0.07, 0.12]	−0.037 [−0.10, +0.03]
W-GNAN $g(w) = w$	0.565 [0.50, 0.63]	0.094 [0.07, 0.13]	−0.024 [−0.08, +0.04]
W-GNAN learned g	0.584 [0.52, 0.65]	0.097 [0.07, 0.14]	−0.005 [−0.07, +0.07]

Table 2: Held-out performance (60/869 positives). AUROC/AUPRC are repeat means; intervals are 1,000-resample subject-cluster bootstraps. Deltas are paired with LR edges; GNN deltas pair repeat 0 only. AUPRC prevalence is .069. *Repeat 0 only, on low-CV sparsified inputs (§4).

5 Results

5.1 Synthetic mechanism check

Each seed generated 700 balanced 36-node graphs (420/140/140 train/validation/test), Beta(2, 2) weights, noise SD 0.12 times latent-score SD, and one topology shared across splits (except the binary case, which varies topology). Graph cases set u_i to a learned centred region constant to isolate the interaction channel. The topology-only model is at chance for the three fixed-topology cases, while weighted variants reach 0.87–0.95 (Table 1). Learned g tracks the generating shape over observed support (Figure 2; correlations .969–.996). Its threshold-case advantage over fixed $g(w) = w$ is descriptive (.95 ± .04 vs. .91 ± .01), not a formal superiority test. With binary weights, all variants coincide within numerical precision, and node-only W-GNAN retrieves the four planted regions with top-4 precision .95.

5.2 Probable-PTSD benchmark

LR edges exceeds the constant-score majority reference in the paired comparison ($\Delta = .087$ [.02, .15]) and reaches AUROC .588 [.52, .65] (Table 2). W-GNAN learned g reaches .584 [.52, .65], with paired $\Delta = -.005$ [−.07, .07]; this interval neither resolves a difference nor establishes equivalence. The fixed-weight and topology ablations

Classification ($PCL-5 \geq 33$)		
Model	AUROC [95% CI]	Δ AUROC vs. LR edges
LR: scale only (total, density)	0.487 [0.42, 0.55]	−0.101 [−0.183, −0.004]
LR: age + sex	0.452 [0.40, 0.50]	−0.135 [−0.21, −0.06]
LR edges, old log1p pipeline	0.595 [0.52, 0.66]	+0.006 [−0.03, +0.04]
LR edges, age/sex/total removed	0.583 [0.52, 0.65]	−0.005 [−0.02, +0.01]

Regression on PCL-5 score [†]		
Model	MAE	R^2 / Spearman ρ
Constant (predict 3)	8.09	—
Ridge on edges	9.40	−0.000 / 0.031 ($p = 0.35$)

Table 3: Selected checks. Classification uses the final repeated-CV protocol. [†]Regression is from an earlier five-fold pipeline and is not directly comparable with Table 2.

are lower but their paired intervals include zero, so they do not establish that weighting is irrelevant. GAT (.539) and GCN (.515) reflect only the stated sparsification and one-repeat design.

5.3 Diagnostics and selected nuisances

Age/sex and total-streamline/density models have AUROC .452 and .487; train-fold residualisation on age, sex and total streamlines changes LR-edge AUROC by −.005 [−.02, .01]. These checks find no predictive contribution from those measured variables under the tested linear controls; they do not rule out all confounding or nonlinear interactions. Exploratory cutoffs 30/25/20 yield strength-LR AUROC .53–.56 (separate five-fold CV; all intervals include .5). Earlier continuous-score ridge regression is worse than a constant in MAE (9.40 vs. 8.09). Across the 15 W-GNAN folds, λ stays near its initial 1 (median .97) and g near identity (Figure 2, right). This is an optimization diagnostic, not proof that the cohort lacks weight-dependent signal.

5.4 Region-level interpretability audit

The earlier prototype’s all-subject seed rankings agree strongly ($\rho = .88$), whereas rankings from its held-out fits agree weakly ($\rho = .21$; Figure 3a). Because architecture, training population and sample size also differ, this contrast demonstrates non-replication in that pipeline but does not isolate subject reuse as the cause. For the final audit (Figure 3b), stability does not exceed its label-permutation null (LR $p = .07$; W-GNAN $p = .85$), and no hypothesis-motivated network crosses $\alpha = .0167$ (Control/Salience/Default: LR $p = .10/.16/.88$; W-GNAN $p = .22/.56/.67$). Resolution is .002 for LR and .010 for W-GNAN; non-significance is not equivalence or evidence of no biological effect.

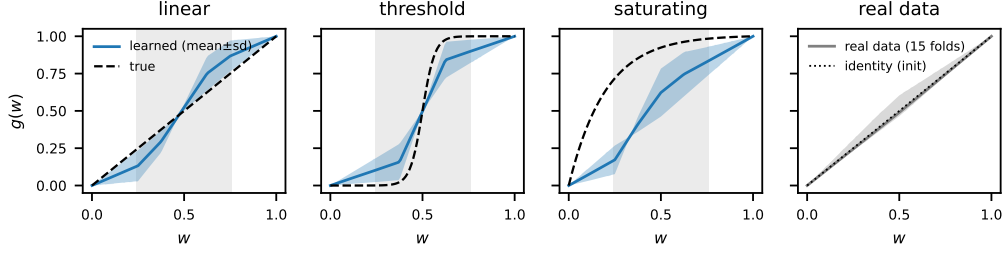


Figure 2: Learned g (mean $\pm$ SD across five seeds) against the generator; grey shading marks the observed-weight support on which correlation is reported. Right: real-data fold curves remain near their identity initialization, a training diagnostic rather than evidence that no nonlinear signal exists.

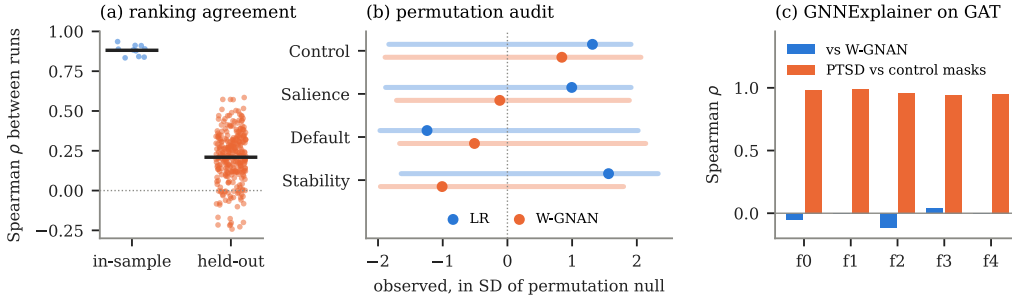


Figure 3: Region-level audit. (a) Mean pairwise Spearman correlation between rankings from five all-subject seeds of an earlier prototype and from its 25 held-out fits. (b) Observed statistics standardized by their own permutation-null mean and SD; bars show null 95% intervals. (c) Per-fold Spearman ρ of positive-subject GNNExplainer node masks with $|t_i|$ group differences (blue) and with control-subject masks (orange).

GNNExplainer’s nonnegative node masks are compared per fold with absolute W-GNAN region allocations (Figure 3c): mean Spearman $\rho = -.03$ (range $-.12$ to $.04$) and top-30 overlap 1.4 regions (chance 2.1, continuous masks ranked by deterministic order). Positive/control masks are nearly identical ($\rho = .96$) and positive-mask stability across folds is $.94$; the explained-set GAT AUROC is $.48$. Mask stability therefore does not establish label-specific information, and mask sign/direction is not recoverable.

6 Discussion

W-GNAN adds a monotone weighted interaction to a specified one-hop GNAN while preserving an exact fixed-graph logit identity. Synthetic controls verify that this channel can use weights when topology is uninformative. In the pilot cohort, W-GNAN and edge-wise LR have an unresolved performance difference; this neither establishes equivalence nor validates a brain mechanism. The region audit similarly finds no network statistic beyond its permutation null. The modest AUROC of about $.58$ matches the limited out-of-sample accuracy of the

largest multisite PTSD classification study (Zhu et al., 2023), and the held-out non-replication of region rankings fits the large cross-validation error bars expected at this sample size (Varoquaux, 2018) and the unestablished status of data-driven PTSD markers (Neria et al., 2022). The nearly label-invariant GNNExplainer masks show that a stable post-hoc explanation need not reflect label-relevant reasoning (Ying et al., 2019; Adebayo et al., 2018); an exact intrinsic decomposition avoids that gap but does not by itself guarantee replicable regions.

Limitations & Deviations. The single-site cohort yields only 60 positive screens and lacks acquisition descriptors. Preprocessing and architecture choices—such as fixed clipping, dropping low-variance edges for GNNs, and omitting contrastive pretraining or multi-head GCNs due to time constraints—restrict the scope. Our GCN/GAT baselines are not exact reproductions of prior work. The audit analyzes region terms t_i rather than pairwise C_{ij} or full perturbations, and permutation budgets differ between models. Future work should validate across sites, retain raw edge support and audit held-out tracts.

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